Experiment-2

Aim: Convert the ER diagram into relational schema. Write complete my SQL create Table statements with primary keys, foreign keys, NOT NULL, Unique, on Delete CASCADE and SET NULL Constraints. Insert Sample data and demonstrate referential Integrity Violation

1) Relational Schema

CUSTOMER(
CustomerID PK,
FirstName,
MiddleName,
LastName,
Email UNIQUE,
DOB
)

CUSTOMER_Phone(
~~CUSTOMER_Phone~~
CustomerID PK/FK,
Phone PK
)

Address (
AddressID PK,
HouseNo,
Street,
City,
State,
PINCode,
CustomerID FK
)

SELLER(
SellerID PK,
SellerName,
Email UNIQUE,
Phone UNIQE,
GSTIN UNIQE,
)

Category (
CategoryID PK,
CategoryName UNIQUE
)

PRODUCT (
PRODUCT ID PK,
PRODuctName,
Price;
StockQuantity,
DEScription,
CategoryID FK
)

Seller_PRODUCT (
SellerID PK/FK,
ProductID PK/FK
)

ORDER (
OrderID PK,
OrderStatus,
Order Date,
TotalAmount,
CustomerID FK
)

ORDER_Item (
orderID PK/FK,
ProductID PK/FK,
Quantity,
UnitPrice,
)

PAYMENT (
PaymentID PK,
PaymentDate,
Amount,

Payment Method,
Payment Status,
TransactionID UNIQUE,
OrderID FK
)

DELIVERY (
DELIVERYID PK,
DeliveryDate,
DeliveryStatus,
TrackingNumber UNIQUE,
OrderID FK,
AddressID FK
)

ELECTRONICS (
ProductID PK/FK
Brand,
Model,
WarrantyPeriod
)

Clothing (
ProductID PK/FK,
Size,
Color;
Fabric
)

GROCERY (
ProductID PK/FK,
ExpiryDate,
Weight,
Brand
)

MySQL Create Table Statements

```
CREATE DataBase IF Not EXISTS IndianCommerce;

Use IndianCommerce;
```

--1. CUSTOMER

```
CREATE TABLE Customer(
CustomerID INT PRIMARY KEY,
FirstName VARCHAR(50) NOT NULL,
MiddleName VARCHAR(50),
LASTName VARCHAR(50) NOT NULL,
Gmail VARCHAR(50) NOT NULL UNIQUE
);
```

-2. CUSTOMER_Phone

```
CREATE TABLE CustomerPhone(
CustomerID INT NOT NULL,
Phone VARCHAR(15) NOT NULL,
PRIMARY KEY (CustomerID, Phone),
FOREIGN KEY (CUSTOMID)
  REFERENCES Customer(CustomerID)
    ON DELETE CASCADE
  );
```

3. Address

```
CREATE TABLE Address(
AddressID INT PRIMARY KEY,
HOUSENO VARCHAR(20) NOT NULL,
Street VARCHAR(100) NOT NULL,
City VARCHAR(50) NOT NULL,
State VARCHAR(50) NOT NULL,
PINCODE VARCHAR(10) NOT NULL,

CUSTOMERID INT NOT NULL,
FOREIGN KEY(CUSTOMERID)
REFERENCES CUSTOMER(CUSTOMERID)
  ON DELETE CASCADE
);
```

4. SELLER

```
CREATE TABLE SELLER (
SellerID INT PRIMARY KEY,
SellerNAME VARCHAR(100) NOTNULL,
Gmail VARCHAR(100) NOT NULL,
Phone VARCHAR(100) NOT NULL,
GSTIN VARCHAR(20) UNIQUE,
);
```

5. CATEGORY

```
CREATE TABLE Category (
CategoryID INT PRIMARY KEY,
CategoryNAME VARCHAR(100) NOT NULL UNIQUE
);
```

6 PRODUCT)

```
CREATE TABLE Product (
    ProductID INT PRIMARY KEY,
    ProductNAME VARCHAR(150) NOT NULL,
    Price DECIMAL(10,2) NOTNULL,
    StockQuantity INT NOT NULL DEFAULT 0,
    Description VARCHAR(500),

    CategoryID INT NOT NULL,

    CONSTRAINT chk_product_price
      CHECK (Price>=0),

    CONSTRAINT chk_product_stock
       CHECK (StockQuantity >=0),

    FOREIGN KEY (CategoryID)
      REFERENCES Category (CategoryID)
      ON DELETE CASCADE,
);
```

7. Seller_PRODUCT

```
CREATE TABLE SellerProduct(
  SellerID INT NOT NULL,
  ProductID INT NOT NULL,

  PRIMARY KEY(SellerID)
    REFERENCES Seller(SellerID)
    ON DELETE CASCADE,

  FORGIGN KEY (ProductID)
    REFERENCES PRODUCT(PRODUCTID)
     ON DELETE CASCADE.
);
```

8. ORDER

```
CREATE TABLE Order(
  OrderID INT PRIMARY KEY,
  orderDate DATE NOT NULL,
  OrderStatus VARCHAR(30) NOT NULL,
  TOTALAMOUNT DECIMAL(12,2) NOT NULL,

  CUSTOMERID INT NOT NULL,

  CONSTRAINT Chk_order_amount
    CHECK (TOTOLAmount >=0),
```

FOREIGN KEY (CUSTOMERID)
REFERENCES CUSTOMER (CUSTOMERID)
ON DELETE CASCADE
);

-9. ORERITEM
- WEAK ENTITY.

CREATE TABLE OrderItem (
OrderID INT NOT NULL,
ProductID INT NOT NULL,
Quantity INT NOT NULL,
UNITPRICE DECIMAL (10, 2) NOT NULL,
PRIMARY KEY (OrderID, ProductID),
CONSTRAIN CHK_OrderItem_PRICE
CHECK (Quantity > 0),
CONSTRAIN CHK_OrderItem_PRICE
CHECK (UNITPRICE >= 0),
FOREIGN KEY (OrderID)
REFERENCES Order (OrderID)
ON DELETE CASCADE,

FOREIGN KEY (ProductID)
REFERENCES PRODUCT(ProductID)
ON DELETE CASCADE
);

-10. RAYMENT

CREATE TABLE Payment(
PaymentID INT PRIMARY KEY,
Payment Date DATE NOT NULL,
Amount DECIMAL (12,2) NOT NULL,
Paymentmethod VARCHAR(30) NOT NULL,
Paymentstatus VARCHAR(30) NOT NULL UNIQUE,
OrderID INT NOT NULL UNIQUE,
CONSTRAIN Chk_Payment_amount
CHECK (Amount >=0)
PORGIGN KEY (Order_ID)
ON DELETE CASCADE
);

11. Delivery

```
CREATE TABLE Delivery (
 DeliveryID INT PRIMARY KEY,
 DeliveryDate DATE,
 DeliveryStatus VARCHAR(30) NOT NULL,
 TrackingNumber VARCHAR(100) NOT UNIQUE,

 OrderID INT NOT NULL UNIQUE,

 AddressID INT,
 Foreign KEY (OrderID)
  REFRENCES order(OrderID)
  on DELETE CASCADE,

 FOREIGN KEY (AddressID)
 REFRENCES Address(AddressID)
  on DELETE SET NULL
);
```

12. ELECTRONICS

```
CREATE TABLE Electronics (
  ProductID INT Primary Key,
  Brand VARCHAR(100) NOT NULL,
  Model VARCHAR(100) NOT NULL,
  WorrantyPerPod VARCHAR(50),
```

```
    FOREIGN KEY (PRODUCTID)
      REFERENCES Product(ProductID)
       ON DELETE CASCADE
    );
```

---

13. Clothing
Product Specialization

```
CREATE TABLE Clothing(
    ProductID INT PRIMARY Key,
    Size VARCHAR(20) NOT NULL,
    Color VARCHAR(30) NOT NULL,
    Fabric VARCHAR(50),
    FOREIGN KEY (ProductID)
     REFERENCES Product(ProductID)
      ON DELETE CASCADE
);
```

14- GROCERY
-- Product Specialization

```
EER CREATE TABLE GROCERY (
      ProductID INT PRIMARY KEY,
      ExpiryDate DATE NOT NULL,
      Weight DECIMAL(6,2)
      Brand VARCHAR(100)
    FOREIGN KEY (ProductID)
       REFERENCES Product(ProductID)
         ON DELETE CASCADE
      );
```

Insert Sample data

① CUSTOMER

```
INSERT INTO CUSTOMER
(CustomerID, FirstName, MiddleName, LASTNAME, Email, DOB)
VALUES
(101, 'Rahul', 'Kumar', 'Sharma', 'rahul@gmail', '2004-05-12'),
```

② Customer phones numbers

```
INSERT INTO CustomerPhone
(CustomerID, Phone)
Values
(101, '9875121661'),
(101, '8752674151'),
```

③ Address

```
INSERT INTO Address
(AddressID, HOUSENO, Street, City, State, PINCode,
                                    CUSTOMERID)

Values
(201, '12A', 'MG Road', 'Lucknow', 'Uttar Pradesh',
                                '226001', 101),
```

④ Sellers

```
INSERT INTO Seller
(SellerID, SellerName, Email, Phone, GSTIN)
values
(301, 'Tech World India', 'techworld@gmail.com',
        '9676415311', '09ABCDE1234'),
```

⑤ Category

```
INSERT INTO Category
(CategoryID, CategoryName)
 Values
(401, 'ELETRONICS');
```

⑥ PRODUCTS

```
INSERT INTO Product
(ProductID, ProductName, Price, StockQuantity, Description,
 CategoryID)
  values
(501, 'DELL LAPTOP', 65000.00, 20, 'Dell Inspiration
                Laptop', 401),
```

PRODUET Specialization data

Electronics

```
INSERT INTO Electronics
(ProductID, Brand, Model, WarrantyPeriod)
  Values
(501, 'DELL', 'Inspiration 15', '2 years');
```

Clothing

```
INSERT INTO Clothing
(ProductID, Size, Color, Fabric)
  Values
(501, 'L', 'Blue', 'Cotton');
```

Grocery

```
INSERT INTO Grocery
(ProductID, ExpiryDate, Weight, Brand)
 Values
(503, '2027-08-15', 5.0g 'Indra Gate');
```

Seller-Product data

```
INSERT INTO SellerProduct
(SellerID, ProductID)
Values
(301, 501)
(302, 502)
(303, 503)
(301, 502)
```

- Seller 301 sells product 501 and 502
- Product 502 is sold by sellers 301 and 302,

Order Data

```
INSERT INTO Order
(OrderID, OrderDate, OrderStatus, TotalAmount, CustomerID)
 values
(601, '2026-06-18', 'Confirmed', 65799.9101),
```

OrderItem Data

```
(OrderID, ProductID)
INSERT INTO OrderItem
(OrderID, ProductID, Quantity, UnitPrice)
  values
(601, 501, 1, 65000.00),
```

Payment data

```
INSERT INTO Payment
(PaymentID, PaymentDate, Amount, PaymentMethod, PaymentStatus
    TransactionID, OrderID)
Values
(701, '2026-06-18', 65799.00, 'UPI', 'Successful',
  'TXN1001', 6001),
```

Delivery data

```
INSERT INTO Delivery
(DeliveryID, DeliveryDate, DeliveryStatus, TrackingNumber,
 OrderID, AddressID)
values
(801, '2026-08-21', 'Out of Delivery', 'Tracking
  Number', 'TRK100001', 600, 201),
```

Demostrate ON DELETE CASCADE

```
DELETE FROM customer
  WHERE CustomerID = 101,

FOREIGN KEY(CustomerID)
REFERENCES Customer(CustomerID)
 ON DELETE CASCADE
```

Demonstrate ON DELETE SET NULL

```
SELECT *
FROM DelRvery
WHERE DelRveryID = 801;
DELETE FROM Address
  WHERE AddressID = 201;
  FOREIGN
```

Referential integrity violation

```
INSERT INTO order
(OrderID, OrderDate, OrderStatus, Total Amount,
    CUSTOMERID)
VALUES
(603, '2026-08-20', 'confirmed', 5000.0, 99);
```

OUTPUT TABLE

```
SELECT
    C.CustomerID,
    CONCAT(C.FirstName,' ',C.LastName) AS
      CUSTOMERNAME,
    C.mail

    O.OrderID,
    O.OrderDate,
    O.OrderStatus,
    O.TotalAmount,
     p.ProductID,
     p.ProductName,
     OI.Quantity,
     OI.UnitPrice,

     S.SellerID,
     S.SellerName,

     Cat.CategoryID,
     Cat.CategoryName,

     pay.PaymentID,
     Cat Pay.PaymentMethod,
      Pay.PaymentStatus,
```

d.DelQueryID,
d.DeliveryStatus,
d.TrackingNumber,

a.City,
a.State,
a.PINcode

FROM Customer C

JOIN 'Order' O
ON C.CustomerID = O.CustomerID

JOIN OrderItem OI
ON O.OrderID = OI.OrderID

JOIN Product P
ON OI.ProductID = P.ProductID

JOIN SellerProduct SP
ON P.ProductID = SP.ProductID

JOIN Sellers
ON SP.SellerID = S.SellerID

JOIN Category Cat
ON P.CategoryID = Cat.CategoryID.

```
LEFT. Join Payment pay
  ON O. OrderID = pay. OrderID

LEFT JOIN Delivery d
  ON O. OrderID = d. OrderID

LEFT JOIN Address a
  ON d. AddressID = a. AddressID;
```

CustomerID	CustomerName	OrderID	ProductName
101	Rahul Sharma	601	Dell Laptop
101	Rahul Sharma	601	Cotton T-Shirt
101	Rahul Sharma	601	Cotton T-Shirt

Customer Id	Customer Name	Order ID	Product Name	Quantity	Seller Name	Category Name	Payment Status	Delivery Status	City
101	Rahul Sharma	601	Dell Laptop	1	Tech World IndPa	ElectronPcs	Successful	out for delivery	Lucknow
101	Rahul Sharma	601	Cotton T-Shirt	1	FashRon Hub	Clothing	Successful	out for delivery	Lucknow
101	Rahul Sharma	601	Cotten T-SHRT	1	Tech World IndPa	Clothing	Successful	out for delivery	Lucknow
101	Priya	602	Basmati Rice	1	Fresh Mart	Grocery	Successful	shipped	Prayagraj

26/08/25

Output